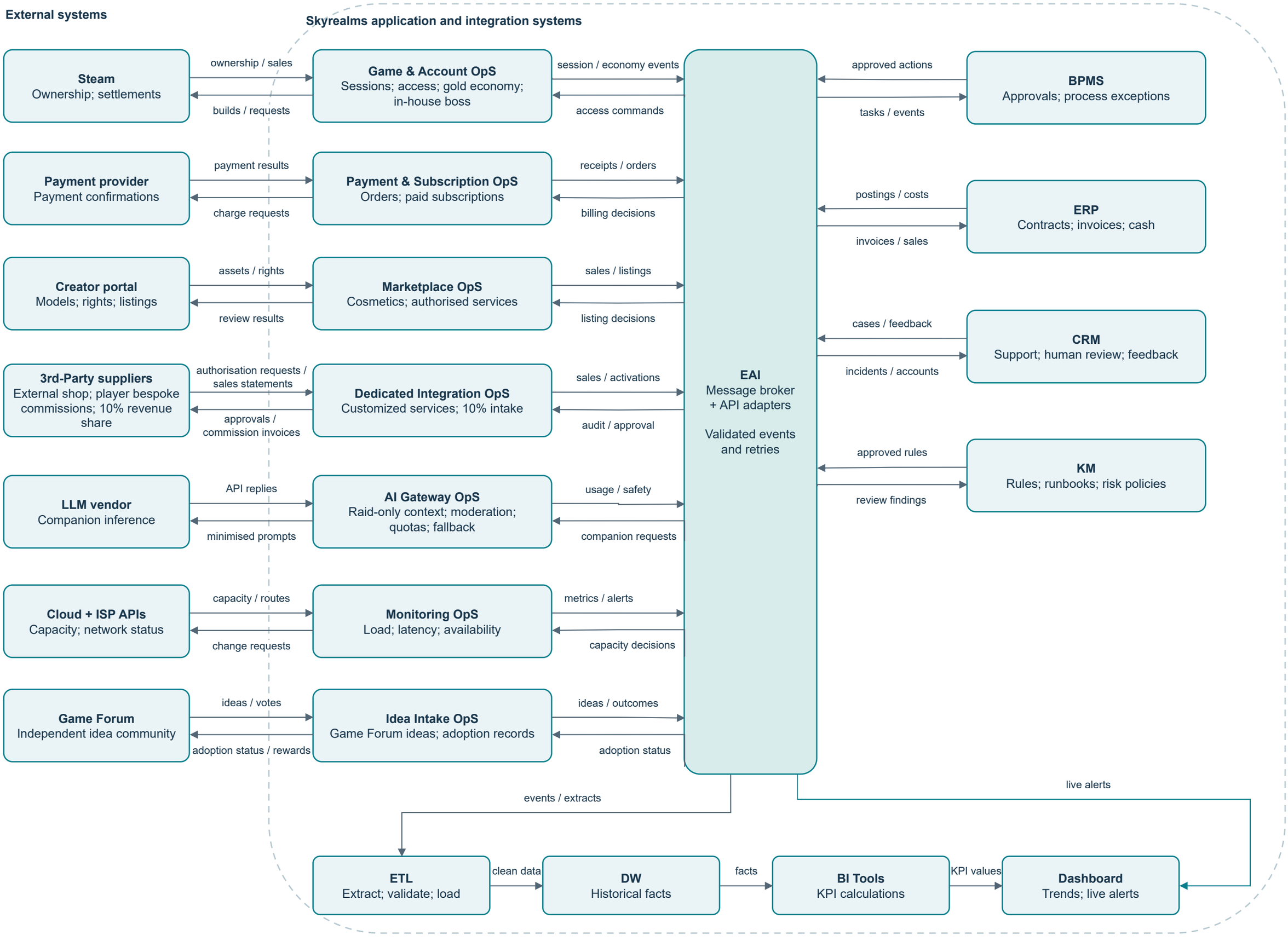
Part 5 | IT Architecture Design

To-be application / integration view | planned launch architecture | labelled, directional data flows



Cloud + ISP APIs are external technology endpoints. The Game Forum is an independent organisation sponsored by Skyrealms. Real-time game packets bypass EAI. Historical KPIs use DW / BI; safety alerts bypass batch ETL.

Part 5 | Business Support Matrix

Adapted from the lecture: systems x business processes. D = direct execution; S = support; - = no planned role. Numbers in the last row are the Section 6 KPIs that use that process.

System / capability	P1 Sales / subscription	P2 Creator procurement	P3 3rd-Party Service Licensing	P4 LLM procurement	P5 Steam release	P6 ISP routing	P7 Cloud capacity	Support / feedback
Game & Account OpS	D	-	-	-	D	-	-	-
Payment & Subscription OpS	D	S	-	-	S	-	-	-
Marketplace OpS	-	D	-	-	-	-	-	-
Dedicated Integration OpS	-	S	D	-	-	-	-	-
AI Gateway OpS	-	-	-	D	-	-	-	-
Monitoring OpS	S	-	-	S	S	D	D	S
BPMS	S	D	D	D	D	D	D	D
ERP	S	D	D	D	D	D	D	-
CRM	S	S	S	-	-	-	-	D
Idea Intake OpS	-	-	-	-	-	-	-	D
KM	S	S	S	S	S	S	S	S
EAI (real-time integration)	S	S	S	S	S	S	S	S
ETL; DW; BI Tools; Dashboard (KPI no.)	S (1-4)	S (4, 6)	S (4)	S (7, 10)	-	S (8, 9)	S (7, 8)	S (5)

Goal support. G1: access, support and retention analysis. G2: reconciled revenue and fair paid offerings. G3: creator review and adopted forum ideas. G4: capacity, network quality and cost control. G5: moderated companions, replaceable LLM access and in-house boss limits.

Weakest dependency. External LLM availability/cost (I7) affects the companion. Use replaceable provider adapters, timeouts, local fallback and quotas (M10); replicate EAI. Vendor resilience alone does not control harmful outputs or hidden incentives.

Risk pattern applied to Skyrealms	Control and architectural responsibility
M: plausible but false game advice / biased replies	AI Gateway OpS restricts context to approved raid facts; KM stores validated rules; CRM routes reports to human review. Keep boss limits outside the LLM.
U: players induce harassment or deceptive output	AI Gateway OpS filters input/output, rate-limits abuse and records minimal evidence; EAI sends urgent alerts directly to Dashboard; CRM supports appeal/review.
H: engagement or data use conflicts with player purpose	Avoid companion pressure to prolong play/spend; make its AI identity clear. Use raid-only memory, short retention and no vendor training/reuse by default; enforce supplier terms and deletion controls.

Risk interpretation. U = undetermined misuse; M = undetermined malfunction; H = hidden goal. These are possible design risks, not claims about a supplier. Their overlap means a working, available system may still harm player interests. Matrix scope is proposed support, not proof of actual deployment or absence of redundant systems.

Basis: Performance Management slides 66/70 (architecture); Risk Patterns slides 34-35, 60-63 (conditions, attachment and privacy). Video lectures inform the To-be view, support matrix and data-quality analysis. Architecture owner keeps interfaces and this model versioned after releases.

Part 6 | Process Measurement

6.1 Leading and Lagging KPIs | required indicator fields from Performance Management slides 71-72

KPI / goal / type	Definition and calculation	Measured process / activity	Target / review / owner	Source IT system	IT solution feeding Dashboard
1. Monthly active licensed players G1.4 / O4 | Lagging	Unique accounts holding a paid base-game licence with >=1 gameplay session in the month; deduplicate Steam/website IDs.	P1 Sales / subscriptions: activate paid access and record gameplay sessions.	>=170,000 by M12 Monthly **Growth lead**	Game & Account OpS	ETL + DW/BI (batch, indirect) EAI -> ETL -> DW -> BI Tools -> Dashboard
2. Month-3 retention G1.3 / O3 | Lagging	New-player cohort active in M+3 / all new licensed players in M x 100%. January cohort is measured in April.	Live game operations: support continued play after initial activation (extends P1).	>=40% by M12 Monthly; matured cohorts **Product lead**	Game & Account OpS	ETL + DW/BI (batch, indirect) EAI -> ETL -> DW -> BI Tools -> Dashboard
3. Subscription attach rate G2.1 / O5 | Lagging	Monthly active licensed players with paid subscription overlapping a gameplay day / all monthly active licensed players x 100%.	P1: select, activate and renew a paid subscription.	>=30% by M12 Monthly **Commercial lead**	Payment & Subscription OpS + Game & Account OpS	ETL + DW/BI (batch, indirect) EAI -> ETL -> DW -> BI Tools -> Dashboard
4. Paid-offering fairness gate G2.3 / O7 | Leading	Offerings verified to give no paid combat advantage / all paid offerings planned for release x 100%. Include subscriptions and marketplace services.	P1/P2: approve paid benefits and cosmetic listings before release.	100% at launch and every release Pre-release **Economy lead**	Marketplace OpS + Payment & Subscription OpS + Dedicated Integration OpS + BPMS	ETL + DW/BI (batch, indirect) EAI -> ETL -> DW -> BI Tools -> Dashboard
5. Implemented player ideas G3.1 / O9 | Leading	Distinct player ideas implemented and released during the quarter. Merge duplicates; approval alone does not count.	Forum feedback: evaluate an idea, implement it and publish the change.	>=10 per quarter Quarterly **Community lead**	Idea Intake OpS + CRM + BPMS	ETL + DW/BI (batch, indirect) EAI -> ETL -> DW -> BI Tools -> Dashboard

Required template: indicator name, definition, goal, target, measured process/activity, source IT system and IT solution feeding Dashboard. Type, frequency and owner are also provided. All targets are planning assumptions; baselines/actuals are not yet measured. P1-P7 match the EPC; live-game and forum activities are explicit extensions.

Measurement notes: M = months after launch for deadlines. In retention formulas, M denotes the cohort month. The 170,000 target is derived from O8: 50,000 paying monthly players / 30% attach rate = approx. 170,000 active licence holders. Show denominators, data freshness and missing-data status.

Dashboard design: trend + target for each KPI; actual value, sample size, last refresh and owner visible. Use grey for missing/stale data, and drill down by region, cohort or supplier. Do not display a planned target as an achieved green result. Actions: investigate 1-3, block unfair releases (4), prioritise useful ideas (5).

Part 6 | Process Measurement

6.1 KPI Design (6-10 of 10) and 6.2 KPI Evaluation

KPI / goal / type	Definition and calculation	Measured process / activity	Target / review / owner	Source IT system	IT solution feeding Dashboard
6. Creator review time G3.2 / O10 | Leading	P90 business days from complete submission to first approve/reject decision; show open cases older than 7 days.	P2: review 3D model submission and issue an approve/reject decision.	P90 <=7 business days Weekly **Creator operations lead**	BPMS + Marketplace OpS	ETL + DW/BI (batch, indirect) EAI -> ETL -> DW -> BI Tools -> Dashboard
7. Cloud + LLM cost ratio Goal4.1 / O11 | Lagging	Monthly accrued cloud and LLM cost / studio revenue x 100%. Revenue excludes VAT/refunds/platform shares; agency trading uses commission revenue only.	P4/P7: consume LLM/cloud services and reconcile supplier costs.	<=20% by M12 Monthly **Finance lead**	ERP + Monitoring OpS + AI Gateway OpS	ETL + DW/BI (batch, indirect) EAI -> ETL -> DW -> BI Tools -> Dashboard
8. Service availability G4.2 / O12 | Lagging	Successful end-to-end probe minutes / scheduled probe minutes x 100%, per region. Includes planned downtime; missing probes require verification.	P6/P7: monitor service, allocate capacity and restore access.	>=99.5% from launch Live + monthly **Reliability lead**	Monitoring OpS	Live: EAI (real time, direct) History: ETL + DW/BI (batch)
9. Gameplay latency G4.2 / O12 | Lagging	Daily P95 round-trip milliseconds between instrumented clients and assigned game servers, separately by supported region.	P6: diagnose traffic; reroute or rebalance server capacity.	<50 ms from launch Live + daily **Network lead**	Monitoring OpS + Game & Account OpS	Live: EAI (real time, direct) History: ETL + DW/BI (batch)
10. Flagged AI replies G5.3 / O15 | Lagging	Distinct replies flagged by filters or player reports / all generated replies x 100%. Include blocked replies; deduplicate reports.	Live AI operation after P4: generate, moderate and review companion replies.	<0.1% by M6 Live + monthly **AI safety lead**	AI Gateway OpS + CRM	Urgent flags: EAI (real time, direct) Rate: ETL + DW/BI (batch)

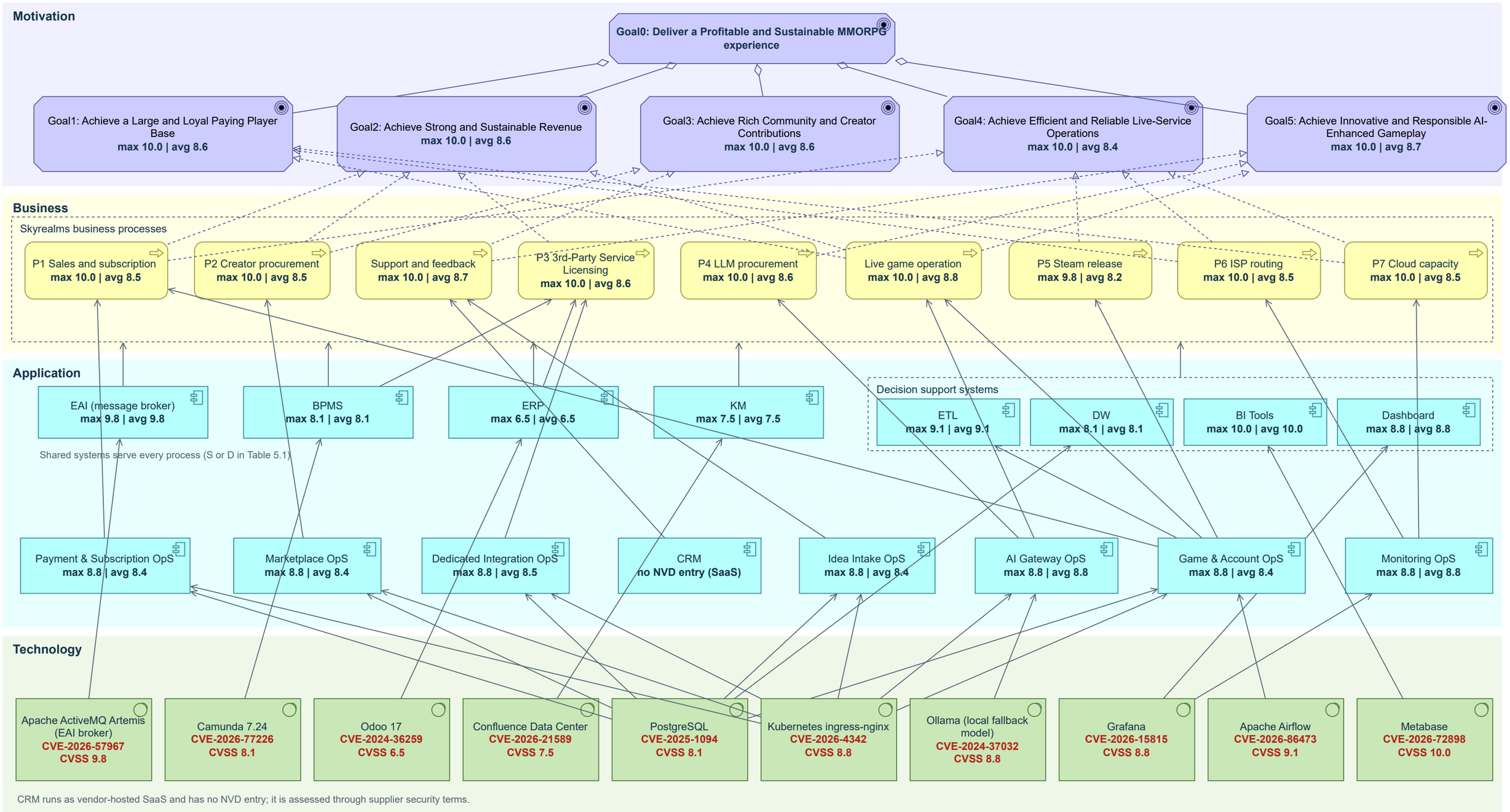
6.2 Evaluation against Eckerson (2006, pp. 200-205). Applied using Performance Management slides 77-78. Three indicators are leading (4-6), seven lagging; leading status is relative to future loyalty/revenue or creator throughput, not to the completion of the measured activity itself. Correlations must be checked, not assumed.

Eckerson criterion	Evaluation and necessary action
Aligned; understandable; few	G1-G5 and measured activities are explicit; there are ten KPIs. Explain P90/P95 and cohort definitions to owners; split out operational diagnostics rather than expanding the executive set.
Balanced; standardised	Read cost (7) together with availability/latency/safety (8-10); never reward cheaper but worse service. Use shared account IDs, business-day calendar, regional scope and revenue definitions for comparison.
Actionable; owned	Owners are named: block unfair offerings (4), improve the idea backlog (5), allocate reviewers (6), tune capacity/quotas (7-9) and investigate AI reports (10). Do not accelerate review by skipping rights/safety checks.
Accurate; timely	Reconcile source events, deduplicate IDs and show denominator/freshness. Urgent incidents use EAI; historical rates use ETL/DW/BI. Missing/stale data is grey and zero denominator is N/A; audit unflagged AI replies.

Critical limits. Retention/subscription growth do not prove player wellbeing. Avoid incentives for compulsive play, emotional dependency or data reuse (risk condition H). A low flag rate can hide under-reporting or over-blocking; sample both allowed/blocked replies. Quarterly review with staff checks whether metrics distort actual work. Remaining diagnostics include companion adoption, boss-limit violations and creator quality.

Part 7 | Enterprise Architecture Model and Vulnerability Impact

ArchiMate layers: motivation, business, application, technology. Technology elements show one reported NVD vulnerability (CVSS 3.1 base score). Scores are propagated upwards with the max and average rules (Ebbers, Hacks and Thakurta, 2021).



Legend. Solid line with open arrow = serving. Dashed line with hollow triangle = realization. Diamond = aggregation (part of). Red text = reported vulnerability from NVD. "max | avg" = propagated CVSS score under the maximum and average rules.